

Akgül et al. 2026 — Rethinking RL for LLM Reasoning

Sparse Policy Selection, Not Capability Learning | arXiv:2605.06241 | [my study repo](#)

One-Liner

RL post-training for math reasoning reranks $\sim 2\%$ of tokens at high-entropy decision points toward alternatives the base model already had in its top-5; an offline contrastive fine-tune gated on base entropy matches full RL at $\sim 1000\times$ less compute.

What’s Novel

- Token-level mechanistic claim with a causal probe: oracle correction at reranked positions recovers RL’s pass@1 exactly; random controls do not.
- Teacher-free locator: base entropy alone ($\tau \approx 1.4$ nats) identifies the intervention sites without ever consulting the RL model.
- Constructive alternative: *ReasonMaxxer* drops the online RL loop, keeps GRPO’s per-group advantage normalisation, applies CE only at decision points, KL-anchors elsewhere.
- RL’s parametric delta is rank-32 LoRA on QKVO (0.27–0.49% of params); rank-8 on W_O alone nearly suffices.

Math in 60 Seconds

Token-level predictive entropy under the base model

$$H_t = - \sum_v \pi_{\text{base}}(v \mid q, o_{<t}) \log \pi_{\text{base}}(v \mid q, o_{<t}).$$

Decision-point set: $D = \{t : H_t > \tau\}$. Per-rollout advantage (from GRPO):

$$A_i = \frac{r_i - \bar{r}}{\sigma_r + \epsilon}, \quad r_i \in \{0, 1\}.$$

ReasonMaxxer loss:

$$\mathcal{L} = - \sum_i A_i \sum_{t \in D^{(i)}} \log \pi_{\theta}(o_t^{(i)} \mid q, o_{<t}^{(i)}) + \lambda \sum_{t \notin D^{(i)}} \text{KL}(\pi_{\text{base}} \parallel \pi_{\theta}).$$

Empirical: reranked fraction 1–4%; the RL-promoted token sits at mean rank ≈ 2.2 under the base; entropy at reranked positions is 5–12 $\times$ that at unchanged ones.

What I’d Push Back On

- “Within base top-5” is measured on already RL-trained checkpoints; partial circularity until verified on a held-out, truly untouched base.
- Threshold τ is tuned per setup. “No RL needed” partially trades the optimisation loop for a scalar grid-search.
- All wins are on math with verifiable scalar rewards. Open-ended tasks, agentic tool use, or tasks where the right intervention is *low*-entropy fall outside the licensed scope — and the framework predicts its own failure there.
- “1000 $\times$ cheaper” compares to full RL training only; ReasonMaxxer still needs labeled-correctness rollouts. Accounting could be sharper.

My One Question

Is the mean-rank- ≈ 2 regularity an artifact of bounded RL learning rates inducing a Lipschitz constraint on rank displacement, or is there something deeper about why RL’s correction never reaches past the second-most-likely base candidate?